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Lead halide coordination competition at buried interfaces for low V_{OC}-deficits in wide-bandgap perovskite solar cells†

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Perovskite tandem solar cells (TSCs) have promising prospects to break the efficiency limit of their single-junction solar cells. Wide-bandgap (WBG) perovskite sub cells as an essential component of perovskite TSCs face challenges in achieving high power conversion efficiency due to severe non-radiative charge recombination. Here, we propose an interfacial engineering strategy of sequentially depositing [2-(9*H*-carbazol-9-yl)ethyl]phosphonic acid and [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid as a hole selective layer, which promotes hole extraction and suppresses the non-radiative recombination at the interface and in the bulk of the perovskite layer. The optimized WBG cells exhibit a remarkable V_{OC} of 1.36 V, resulting in a record low V_{OC} deficit of 0.42 V. Moreover, integrating these WBG perovskite sub cells into all-perovskite tandem cells gives rise to outstanding power conversion efficiencies of 27.34% and 28.05% for two-terminal and four-terminal TSCs, respectively. This research provides a unique perspective for the innovative preparation of highly efficient halide perovskites with high-quality buried interfaces.

Broader context

Tandem solar cells (TSCs) have been showing great application prospects due to their potential to break the Shockley–Queisser (S–Q) limit of single-junction solar cells by reducing thermalization losses. However, as the front unit, a wide-bandgap (WBG) cell that collects high-energy photons suffers a severe open-circuit voltage (V_{OC}) deficit mainly due to a large number of defects at the interface between the perovskite and charge transport layers and the poor perovskite crystalline quality, which impedes the further advancement of TSCs. Here, we introduce a simple and straightforward interface engineering strategy using a stacking layer, *i.e.*, 2PACz/Me-4PACz, as the hole selective layer. By controlling the crystallization process of perovskites and charge carrier transport at the buried interface, the non-radiative charge recombination is significantly suppressed together with a great improvement in perovskite crystallization quality, leading to enhanced performance of both single-junction wide-bandgap cells and tandem solar cells. This research enriches our understanding of perovskite crystalline growth through bottom interface management and further advances the basic strategy of commercialization of perovskite solar cells.

1. Introduction

Metal halide perovskites have garnered significant attention from both academic and industrial communities in the past decade due to their exceptional optoelectronic properties.[1–5] Notably, power conversion efficiency of single-junction perovskite solar cells (PSCs) has been skyrocketing to 26.1%.[1,6–9] To break the efficiency limit of single-junction cells, perovskite tandem solar cells have been developed rapidly in the last several years.[10–22] Among them, all-perovskite tandem solar cells have promising prospects, thanks to their adjustable bandgaps ranging from 1.1 to 2.3 eV. So far, the highest efficiency achieved for all-perovskite tandem solar cells has been 28%, surpassing that of single-junction ones.[22] However, front wide-bandgap (WBG) cells suffer severe open-circuit voltage (V_{OC}) deficit, which greatly hinders the advancement of perovskite tandem solar cells.[23–27] The V_{OC} deficit of WBG PSCs ($\geq$0.5 V) is unprecedentedly much larger than that of conventional bandgap and narrow-bandgap (NBG) solar cells ($\sim$0.3 V). This is mainly due to the following aspects: (1) non-radiative recombination caused by defects and energy level mismatch at the interface between WBG perovskites and charge selective

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layers;[10,23,28,29] (2) WBG perovskites have large bulk defects due to their fast crystallization rate;[14] and (3) the optoelectronic characteristics of hole selective layers and their influence on perovskite crystallization.

Regarding the first two aspects, much effort has been made.[10,14,23,24,30–37] For example, Sargent *et al.* reduced the V_{OC} deficit by suppressing non-radiative recombination at the perovskite/C_{60} interface through 1,3-propane diammonium (PDA) post-treatment.[10] Zhao *et al.* reduced the V_{OC} deficit at the interface by using self-assembled monolayers (SAMs) and 2-thiopheneethylammonium chloride (TEACl) post-processing.[36] Zhu *et al.* prepared a WBG perovskite layer by gas quenching to obtain a highly crystalline perovskite layer.[14] However, the study of the hole selective layer remains unsatisfactory.

At present, the hole selective layers most commonly used in WBG PSCs are mainly as follows: (1) NiO_X hole selective layer: its low-temperature process, strong chemical stability, high optical transmittance, and low manufacturing cost have attracted broad interest but it suffers from substantial defects, resulting in a large V_{OC} deficit[38–40] and (2) self-assembled monolayer: represented by [2-(9*H*-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz). But they have their own shortcomings. 2PACz has a weaker adsorption capacity to the ITO surface, while the high hydrophobicity of Me-4PACz makes achieving full perovskite coverage difficult.[13,27,41–44] In order to solve these issues, several methods have been proposed to optimize the process of the hole selective layer, combine the use of different hole selective layers, or develop novel hole selective materials.[12,20,42,43,45–47] However, it is difficult to simultaneously achieve high performance in hole extraction and the regulation of the crystallization of the perovskite layer. For example, Liu *et al.* deployed NiO_X/2PACz as a hole selective layer. The NiO_X intermediate layer helps to uniformly self-assemble 2PACz molecules onto substrates, thereby avoiding direct contact between ITO and perovskites, minimizing the shunt loss, mitigating the V_{OC} deficit, and improving the device performance.[42] Sargent *et al.* spun cast Me-4PACz on NiO_X to address the surface defects of NiO_X and the non-wetting issue associated with Me-4PACz.[10] Ahmed *et al.* reduced interfacial loss by evaporating SAM HTLs.[43] Tan *et al.* used a hybrid layer by mixing two hole-selective molecules to form SAMs and bridged perovskites with low-temperature treated NiO_X nanocrystalline films. This approach enables the hole selective layer to energetically match with the perovskite absorber to reduce the V_{OC} deficit.[48]

Here, we present a method for interface engineering and investigate its influence on device performance, particularly on V_{OC} and the fill factor (FF). The method involves using SAM 2PACz as the initial hole selective layer and subsequently modifying it with Me-4PACz. Characterization results reveal that interface engineering with Me-4PACz not only alters the surface potential distribution of 2PACz on the ITO surface but also influences the crystallization of the subsequent perovskite layer. This leads to a denser perovskite film at the buried interface, resulting in an improved energy level alignment between the perovskite layer and the hole selective layer and suppressed non-radiative recombination. We therefore achieve a champion PCE of 19.83% and a V_{OC} of 1.36 V in a PSC with a 1.78 eV bandgap, which represents one of the lowest V_{OC} deficits (0.42 V) among WBG PSCs. Notably, the strategy enables the realization of a 2-T all-perovskite tandem solar cell with an impressive efficiency of 27.34% and a 4-T all-perovskite tandem solar cell with a champion efficiency of 28.05%. This provides guidance for the further development of WBG perovskite solar cells and tandem solar cells.

2. Results and discussion

The Me-4PACz interface engineering process is illustrated in Fig. 1a. For clarity in subsequent discussions, we denote perovskite films or devices without Me-4PACz interface engineering as the control, and those with Me-4PACz interface engineering as the target. To investigate whether Me-4PACz is deposited onto the 2PACz monolayer, we performed the contact angle measurement (Fig. S1, ESI†). The contact angle shows a significant change. The control sample displays a contact angle of 13.2°, while the target one shows a contact angle of 23.6°. These findings indicate the successful deposition of Me-4PACz molecules onto the 2PACz monolayer. We further examine the evolution of P 2p peaks on different substrates through X-ray photoelectron spectroscopy (XPS) measurement to ensure that they are not washed away during the subsequent perovskite deposition (Fig. S2, ESI†).

To investigate the effect of Me-4PACz interface engineering onto the 2PACz monolayer, we employed Kelvin probe force microscopy (KPFM) to examine the surface potential distribution in the control and target samples (Fig. S3, ESI†). As anticipated, the surface potential distribution after Me-4PACz interface engineering appears to be more uniform. Moreover, the modified interface demonstrates a deeper highest occupied molecular orbital (HOMO) level compared to 2PACz.

We deposited perovskite solution onto different substrates and recorded the formation of perovskite films (Fig. S4, ESI†). The images show that the perovskite film can be readily deposited onto 2PACz, while it is difficult to deposit onto Me-4PACz.[27] However, to our surprise, the perovskite film fully covers the 2PACz/Me-4PACz substrates. This might be due to the fact that the filling of the 2PACz vacancy with an appropriate amount of Me-4PACz leads to an increase in the contact angle with the perovskite solution. However, the remaining Me-4PACz tends to accumulate on the surface rather than forming dimers with 2PACz molecules due to the positive binding energy value (∼0.205 eV) and excessive spatial distance (∼3.79 Å) from 2PACz molecules (Fig. S5, ESI†). Additionally, the phosphate group is hydrophilic, which eases the difficulty of perovskite solution deposition.[9,49]

Photoluminescence quantum yield (PLQY) is widely employed to quantify the degree of non-radiative recombination in perovskite films. We spin-cast WBG perovskite $(FA_{0.8}Cs_{0.2})Pb(I_{0.6}Br_{0.4})_3$ films on various substrates and

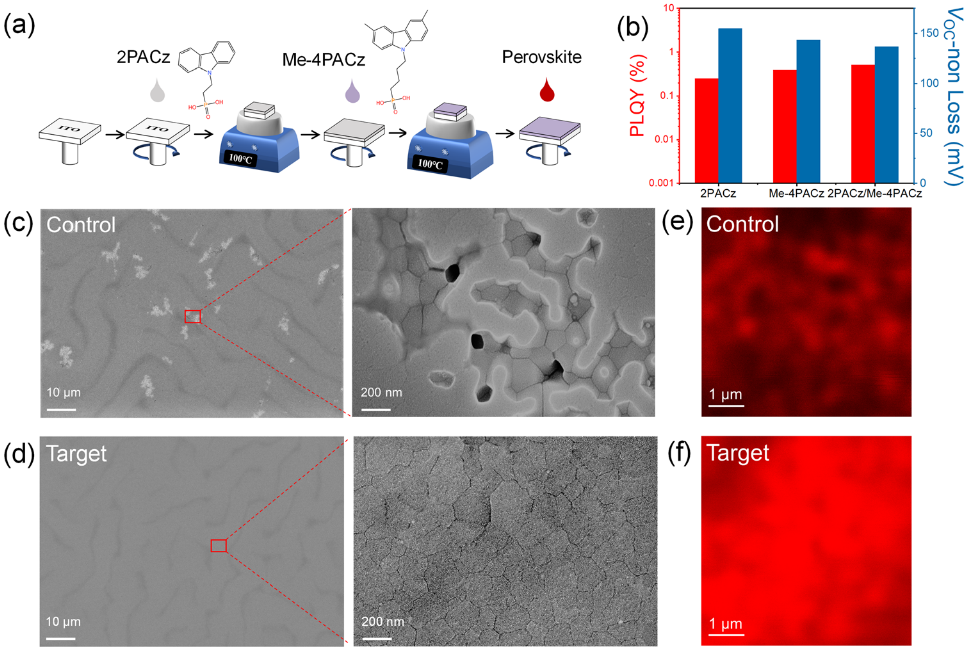

Fig. 1 (a) Schematic diagram of the Me-4PACz interface engineering process. (b) PLQY values of perovskite films deposited on different substrates and the corresponding V_{OC} non-radiative recombination loss calculated from the PLQY value. Top-view SEM images of the control (c) and target (d) perovskite films at the perovskite buried interface. PL mapping patterns of the (e) control and (f) target perovskite films at the perovskite buried interface using a 552-nm laser.

measured their corresponding PLQY. The results are shown in Fig. 1b. Clearly, the perovskite film with Me-4PACz interface engineering exhibited a higher PLQY (0.51%) than the control one (0.25%). The improved PLQY suggests a reduction in non-radiative recombination and an enhancement in radiative recombination (Note S1, ESI†),[50] owing to defect passivation at the interface and regulation of perovskite layer growth. The V_{OC} loss due to non-radiative recombination is also further reduced (from 155 to 137 mV).

To investigate the factors contributing to the increased PLQY value, we uncovered the buried interface of the perovskite layer using the method illustrated in Fig. S6 (ESI†) and examined it with scanning electron microscopy (SEM). The control perovskites exhibited numerous spots at the buried interface (Fig. 1c). Upon further inspection, it became apparent that these spots correspond to voids at the interface, potentially containing residual lead iodide that has not undergone complete conversion into perovskite.[51] This phenomenon may be attributed to the uneven distribution of 2PACz molecules on the substrate. Conversely, the buried interface of the target sample exhibits a completely different pattern with a dense perovskite film with enlarged grains (Fig. 1d). These observations suggest that the Me-4PACz interface engineering layer plays a significant role in the crystallization of the buried interface of the perovskite layer.[52]

We examined the potential distribution of the buried perovskite surface (Fig. S7, ESI†). The control sample showed a highly uneven potential distribution, which can be attributed to the non-uniform distribution of the 2PACz monolayer and the detrimental effects of these voids. In contrast, the target sample exhibited a uniform potential distribution on its buried surface, likely resulting from the surface potential distribution facilitated by Me-4PACz interface engineering and hence the growth of perovskite. This uniform potential distribution promotes smooth carrier transport, thereby suppressing non-radiative recombination.[53]

Furthermore, we conducted photoluminescence (PL) mapping of the perovskite buried surface (Fig. 1e and f). It revealed that the target perovskite film exhibited a more uniform PL emission than the control one, indicating the absence of voids. Moreover, the target film demonstrated stronger PL emission, which can be attributed to the passivation effects of Me-4PACz. Both the uniform PL emission and enhanced PL intensity give rise to the reduction of charge non-radiative recombination and efficient charge extraction from the perovskite layer.

Fig. 2a shows the X-ray diffraction (XRD) patterns of the perovskite films deposited on these two substrates with different hole selective layers. They exhibited an identical crystal structure with no additional crystalline peaks. However, they differ quite significantly in the intensity ratio of the (100) peak to the (110) peak (Fig. S8, ESI†). The target perovskite film exhibited a higher intensity ratio of 1.9 than the control (1.3). In the meantime, the full width at half maximum (FWHM) of the diffraction peak corresponding to the (100) plane slightly narrows from 0.22° to 0.21° after Me-4PACz interface engineering.

Additionally, the potential stress of the control and target perovskite films was investigated using grazing incidence X-ray diffraction (GIXRD) (Fig. 2b and c). As the diffraction angle increased from 1° to 3°, the 2θ = 32.4° peak of the control

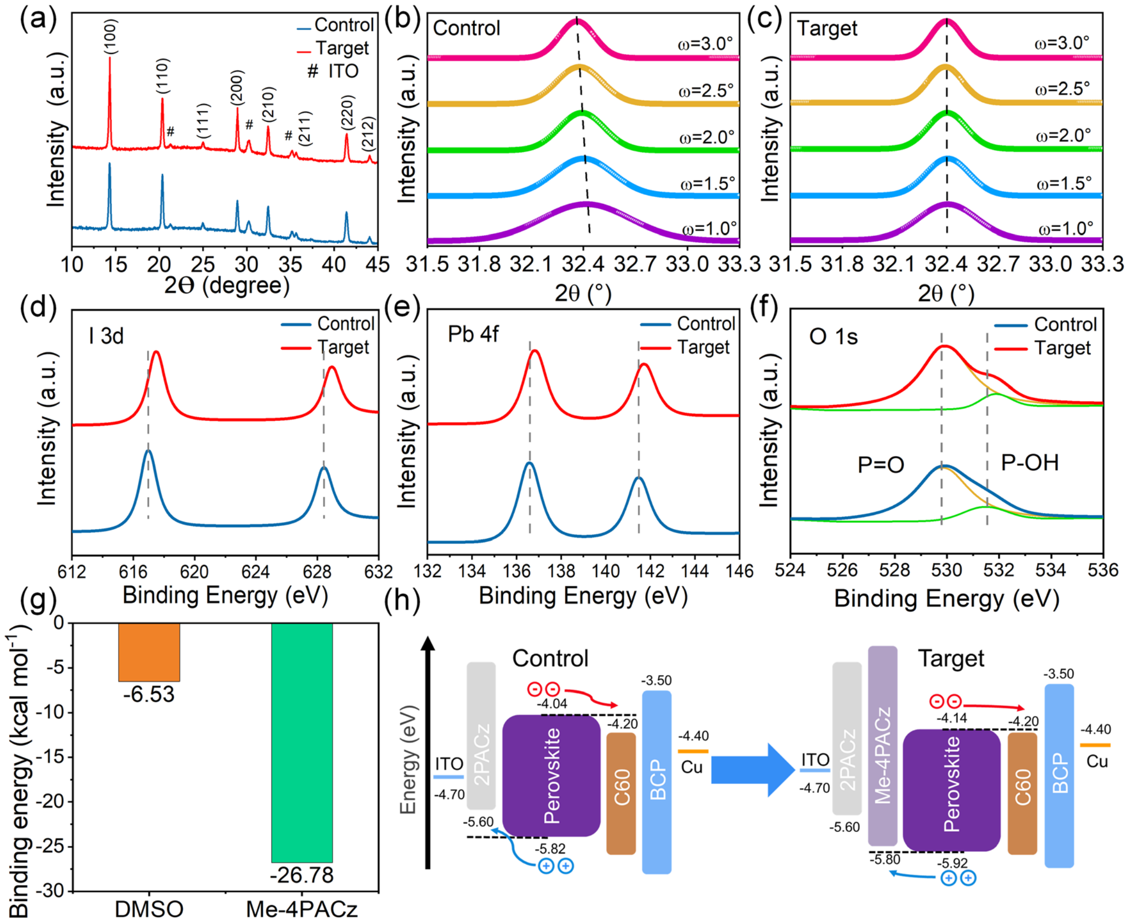

Fig. 2 (a) XRD patterns of these perovskite thin films. GIXRD spectra of the control perovskite film (b) and the target perovskite film (c) measured at different grazing-incidence angles. XPS spectra of I 3d (d), Pb 4f (e) and O 1s (f) in the perovskite films with and without Me-4PACz interface engineering. (g) The binding energies of DMSO molecules and Me-4PACz molecules with PbI_2 molecules. (h) The energy level diagrams for the control and target devices.

perovskite film gradually shifted to a smaller angle, which may be due to the formation of voids. In contrast, the peak value of the target film showed negligible change in the diffraction angle, indicating the absence of stress and strain.

The XRD results demonstrate that Me-4PACz interface engineering enhances crystallinity and alleviates stress in the perovskite films. As a result, the improved structural order, reduced lattice defects, and stress relief contribute to enhanced carrier transport and reduced recombination, ultimately leading to improved device performance.[54,55]

We conducted XPS measurement to investigate the potential interaction at the buried interface that may influence the crystallization of the perovskite film. As shown in Fig. 2d–f and Fig. S9 (ESI†), there are apparent shifts in the binding energies of the I 3d, Br 3d, and Pb 4f peaks of the target perovskite film in comparison to the control. Additionally, the peak positions corresponding to the P–OH group (at 531.5 eV) and the P═O group (at 530.0 eV) in the O 1s spectrum also exhibited shifts. Specifically, the shifts in the Pb 4f and O 1s peaks suggest the formation of covalent bonds (P═O–Pb) resulting from the interaction between phosphonates and Pb. Furthermore, the shifts in the P–OH group peak position in the O 1s spectrum indicate the formation of hydrogen bonds (P–OH·I/Br) due to the higher electronegativity of oxygen compared to halogen. These interactions between Me-4PACz and the perovskite provide additional evidence for the influence of Me-4PACz on the crystallization and surface chemistry of the perovskite film, contributing to improved device performance.[56]

To gain a deeper understanding of the interaction mechanism at the buried interface, we calculated the binding energy of the interaction of DMSO and Me-4PACz with PbI_2 (Fig. 2g and Fig. S10, ESI†). The binding energy of DMSO·PbI_2 was calculated to be -6.53 kcal mol^{-1}, while that of Me-4PACz·PbI_2 was to be -26.78 kcal mol^{-1}. This indicates that Me-4PACz exhibits stronger coordination with PbI_2 compared to DMSO.

Ultraviolet photoelectron spectroscopy (UPS) measurements were performed to assess the energy level alignment at the hole or electron transport layer interface (Fig. S11, ESI†). The energy level diagrams for the control and target devices (Fig. 2h) show that the conduction band minimum (CBM) and valence band maximum (VBM) of the perovskite film in the target device are

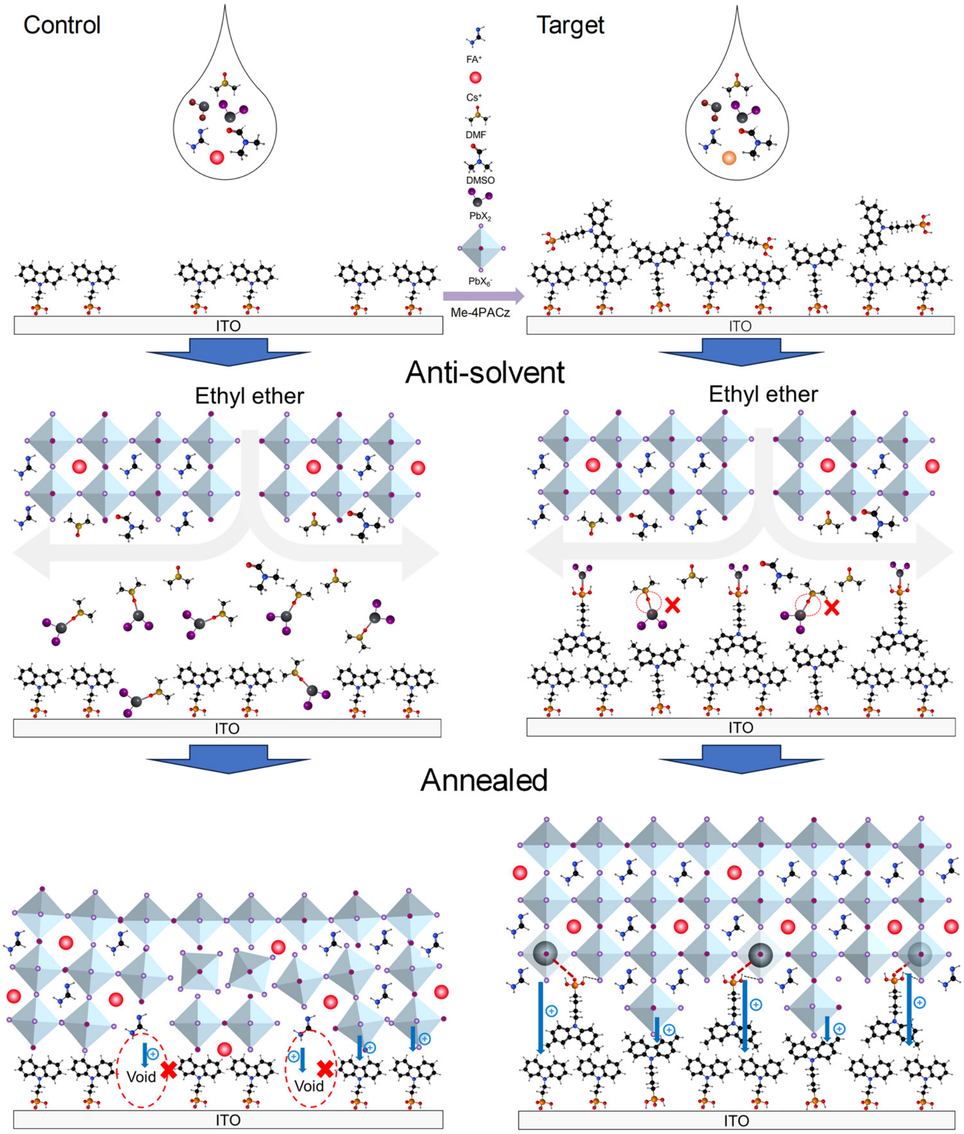

Fig. 3 Schematic diagram of the Me-4PACz interface engineering mechanism. With Me-4PACz interface engineering, the adsorption vacancies caused by 2PACz are filled, and the remaining Me-4PACz molecules compete with DMSO to coordinate with PbI_2, accelerating the dissociation of DMSO and PbI_2 at the buried interface, promoting the crystallization and formation of a dense perovskite layer.

shifted downwards compared to the control device.[41,57] This facilitates electron transport from the perovskite layer to the electron transport layer (C_{60}). Additionally, the insertion of the Me-4PACz layer greatly reduces the energy level mismatch between the hole selective layer and the perovskite layer, enhancing hole transport from the perovskite layer to the hole selective layer while further suppressing electron transport from the perovskite layer to the hole selective layer.

We then propose the following mechanism model (Fig. 3): the weak adsorption ability of 2PACz onto the ITO surface readily results in vacancies and an uneven surface potential distribution. During the anti-solvent drop-casting process, it is inevitable that many DMSO-PbI_2 molecules will be present at the buried interface. Further thermal annealing leads to the formation of voids at the buried interface. However, with Me-4PACz interface engineering, the voids are filled, and the potential distribution at the substrate becomes more uniform. Furthermore, some Me-4PACz molecules accumulate in a disordered manner on the surface. During the anti-solvent drop-casting process, the phosphate group competes with DMSO to coordinate with PbI_2, accelerating the dissociation of DMSO and PbI_2 at the buried interface. Meanwhile, the hydrophobic methyl group and the hydrophilic phosphate group work together to orient the Me-4PACz molecule in an upward phosphate arrangement. As a result, there is less DMSO at the buried interface, promoting the crystallization and formation of a dense perovskite layer.

Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements were conducted to examine the charge dynamics of perovskite films. As shown in

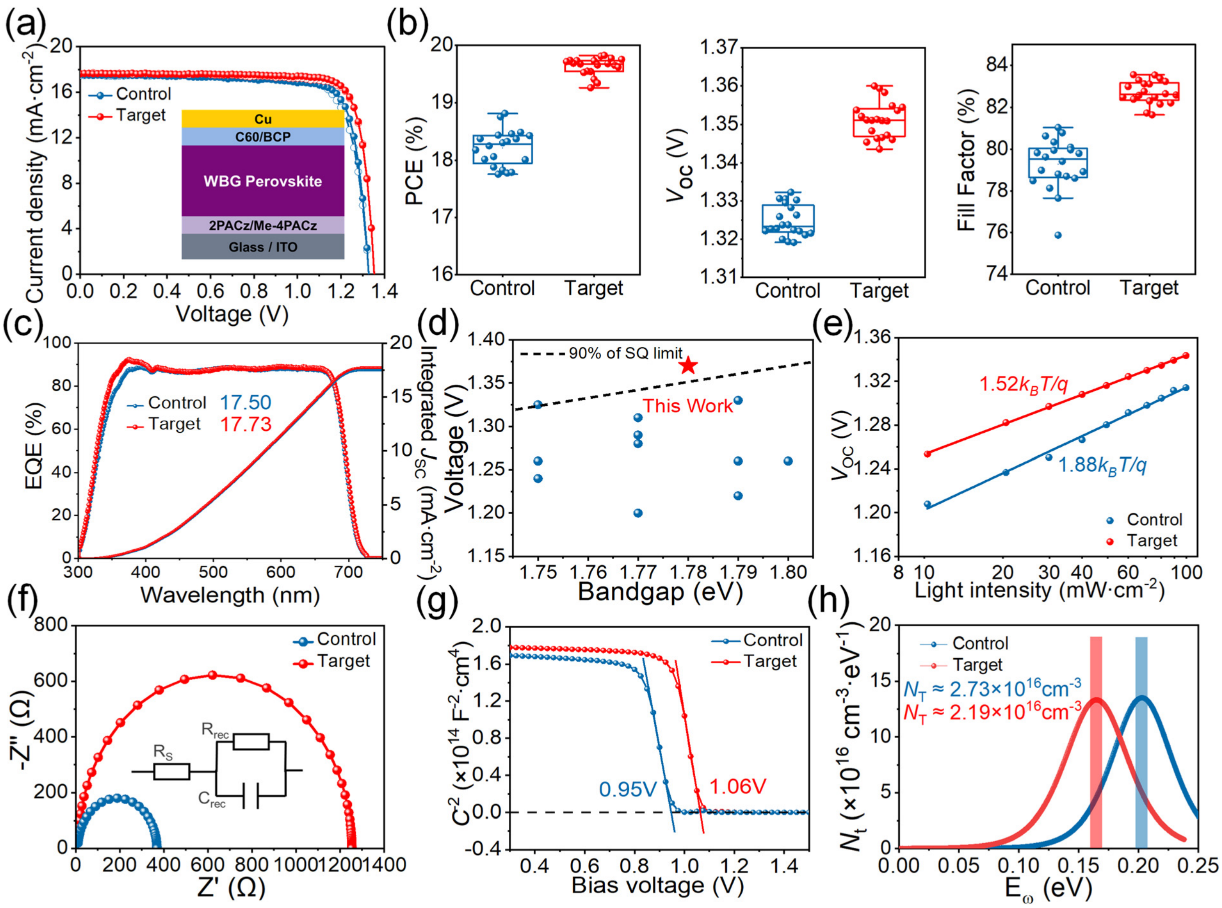

Fig. 4 (a) J–V curves of champion devices in the control group and target group (the illustration is a schematic diagram of the device structure). (b) Box chart of device performance parameters (V_{OC}, FF, and PCE) in the control group and target group. (c) EQE curves of the champion devices in the control group and target group. (d) Reported V_{OC} *versus* bandgap values for wide bandgap perovskite solar cells. The red star represents this work using Me-4PACz interface engineering. The light intensity-dependent J–V curves (e), Nyquist plot (f) and capacitance–voltage curves (g) of control and target devices. (h) The trap state density at room temperature calculated from admittance spectra of control and target devices.

Fig. S15a (ESI†), the PL intensity of the target samples was significantly higher than that of the control, suggesting the enhanced crystallinity of the target perovskite films and hence substantial suppression of non-radiative recombination. This is further supported by the time-resolved photoluminescence (TRPL) spectra shown in Fig. S15b and Table S1 (ESI†). The target perovskite film exhibited a much longer average carrier lifetime of 2257 ns, in contrast to the control one with a lifetime of 872 ns. Light-induced phase segregation is a very common phenomenon for wide bandgap perovskite films. We therefore monitored the temporal evolution of PL spectra of perovskite films without and with Me-4PACz treatment (Fig. S16, ESI†). It can be seen that the light-induced phase segregation in the target sample is significantly suppressed compared to the control sample. This may be due to the interaction of phosphate with halogen, which leads to the reduction of defects.

We fabricated single-junction WBG perovskite solar cells with the device structure of ITO/hole selective layer/perovskite/C_{60}/BCP/Cu, as illustrated in Fig. 4a. We have fabricated 20 independent devices for each variable to ensure reliable statistical analysis. The control cell yielded a champion PCE of 18.76% (18.49%) with a V_{OC} of 1.33 V (1.33 V), short-circuit current density (J_{SC}) of 17.54 mA cm^{-2} (17.47 mA cm^{-2}), and FF of 80.63% (79.80%) under reverse (forward) scan. By contrast, the target cell delivered a champion PCE of 19.83% (19.67%), with a V_{OC} of 1.35 V (1.35 V), J_{SC} of 17.65 mA cm^{-2} (17.62 mA cm^{-2}) and FF of 83.11% (82.64%) under reverse (forward) scan. The stabilized power output of the target cell reached 19.73% (Fig. S17, ESI†), outperforming the control one of 18.61%. The statistical distribution of device metrics is presented in Fig. 4b and Fig. S18 (ESI†). Notably, compared to the control ones, the target cells exhibit significant improvements in V_{OC} and FF, resulting in higher average PCE.

External quantum efficiency (EQE) of the champion cells is illustrated in Fig. 4c. The target cell demonstrated higher EQE values and integral J_{SC} than the control one, consistent with the J–V scan results. Furthermore, we achieved a champion V_{OC} of 1.36 V, which represents one of the highest V_{OC} values and lowest V_{OC}-deficit value reported for WBG (1.75–1.85 eV) perovskite solar cells (Fig. 4d and Fig. S19–S21, Table S2, ESI†).

We performed maximum power point tracking (MPPT) measurements for both the control and target devices, as recorded in Fig. S22 (ESI†). It shows that the efficiency of the target device remains at 80% of the initial PCE after 400 h, while the efficiency of the control device drops to 80% of the initial PCE after 200 h. This enhancement in stability can be primarily attributed to several factors. First, compared to the unevenly distributed and porous perovskite layer formed by 2PACz, the interaction of the phosphate functional group of Me-4PACz with $PbI_2/PbBr_2$ leads to the formation of a dense buried interface. This interface improves the structural stability of the perovskite film by releasing stress and promoting a higher degree of crystallinity. A more crystalline and stable perovskite film is less prone to degradation, leading to improved long-term stability of the device. Furthermore, the reduction of non-radiative recombination centers plays a crucial role in stabilizing the device performance. The Me-4PACz interface engineering effectively suppresses non-radiative recombination processes by reducing trap densities and improving charge carrier dynamics. This improvement allows for more efficient extraction and transport of charge carriers, minimizing energy losses and enhancing the stability of the device during prolonged operation. Overall, the combination of improved structural stability and reduced non-radiative recombination centers provided by the Me-4PACz interface engineering contributes to the enhanced stability observed in the target cells.

To investigate the impact of Me-4PACz interface engineering on charge recombination, we conducted light intensity-dependent *J–V* measurements. As shown in Fig. 4e, V_{OC} exhibits a logarithmic increase with increasing light intensity. The relationship was fitted using eqn (1):

$$V_{OC} = nkT\ln(L)/q \tag{1}$$

where n, k, T, L, and q represent the ideality factor, Boltzmann's constant, absolute temperature, light intensity, and elementary charge, respectively. Compared to the control devices, the target devices showed a reduced n value, decreasing from 1.88 to 1.52. This confirms that Me-4PACz interface engineering effectively suppresses trap-assisted non-radiative recombination losses, resulting in higher V_{OC} and FF in solar cells.

We also conducted electron impedance spectroscopy (EIS) measurements to understand the transport mechanism of WBG devices, and the results are summarized in Table S3 (ESI†). The Nyquist plot (Fig. 4f) was obtained by measuring the control and target devices in dark with a bias voltage of 1.18 V and a sweep frequency range of 10 MHz to 10 Hz. The Me-4PACz interface molecular engineering led to a decrease in series resistance (R_S) from 9.3 to 8.9 Ω and an increase in recombination resistance (R_{rec}) from 364.3 to 1253.4 Ω. Compared to the control ones, the target devices exhibit lower R_S and higher R_{rec} in both low and high-frequency regions, indicating limited charge recombination and better carrier transport.

Capacitance–voltage (*C–V*) measurements were performed to estimate the built-in potential (V_{bi}) using Mott–Schottky analysis, as shown in Fig. 4g. The target devices exhibited a larger V_{bi} of 1.06 V compared to the control devices (0.95 V). The increase in V_{bi} is attributed to the suppression of non-radiative recombination through Me-4PACz interface molecular engineering. The trap state density (N_t) was estimated using thermal admittance spectroscopy (TAS) measurements (Fig. 4h and Fig. S24, ESI†), revealing an N_T of 2.19×10^{16} cm^{-3} for the target perovskite film and 3.73×10^{16} cm^{-3} for the control one.

We also applied the method to other WBG perovskite compositions like 1.66 eV $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$, which is suitable for perovskite/silicon tandem solar cells. The obtained results (Fig. S25, ESI†) exhibit similar trends, indicating the universality of the Me-4PACz interface engineering strategy. This is beneficial for the advancement of tandem solar cell technologies. Comparison studies using different hole selective layers have been carried out, further demonstrating the advantages of our approach (Fig. S26, ESI†). Furthermore, we also deposited the 2PACz/Me-4PACz stacked layer using the blade-coating method for cell fabrication. As shown in Fig. S27 (ESI†), the reverse (forward) scanned efficiency of the champion device reached 19.58% (18.57%), demonstrating the scalability of our strategy.

We then fabricated monolithic all-perovskite tandem solar cells using Me-4PACz interface engineering for the WBG sub-cells (Fig. 5a). The champion tandem device exhibited a reverse (forward) scanned PCE of 27.34% (26.84%) with a V_{OC} of 2.15 V (2.14 V), J_{SC} of 15.81 mA cm^{-2} (15.80 mA cm^{-2}) and FF of 80.49% (79.39%) and a stabilized efficiency of 27.25% measured over 300 s (Fig. 5b and Table 1). The integrated J_{SC} values obtained from EQE spectra for the WBG and NBG sub-cells are 16.0 and 15.5 mA cm^{-2}, respectively, which are in good agreement with the J_{SC} values determined from *J–V* measurements (Fig. 5c). We also fabricated 4-T all-perovskite tandem solar cells combining semi-transparent WBG PSCs as the front sub-cell and NBG PSCs as the bottom cell. The device structures of the semi-transparent WBG PSCs, NBG PSCs, and 4-T tandem cells are shown in Fig. 5d. To protect the WBG perovskite layer during the sputtering of ITO as the transparent top electrode, an atomic layer-deposited (ALD) SnO_X layer (20 nm) was utilized instead of BCP. The *J–V* curves of the semi-transparent WBG PSC, independent NBG PSC, and filtered NBG PSC are shown in Fig. 5e. Compared with the opaque one, the semi-transparent device exhibits lower parameters. The attenuation of light absorption of the perovskite layer caused by the transparent ITO top electrode contributes to the decrease in J_{SC} and V_{OC}, which aligns with the significant decline in EQE spectra within the wavelength range of 530 to 710 nm (Fig. 5f). The decrease in the FF may be attributed to sub-optimal contact interfaces. The EQE integrated J_{SC} of the semi-transparent WBG device aligns well with the J_{SC} obtained from the *J–V* curves.

The semi-transparent WBG PSC demonstrates a champion PCE of 18.25% (17.87%), and the filtered NBG rear cell shows a champion PCE of 9.80% (9.44%) under reverse (forward) scan. As a result, the combination of the front sub-cell and the rear sub-cell yields an all-perovskite tandem cell with an efficiency of 28.05%. The corresponding photovoltaic metrics of the 4-T tandem cell are also summarized in Table 1. This efficiency is one of the highest reported for all-perovskite 4-T tandem solar

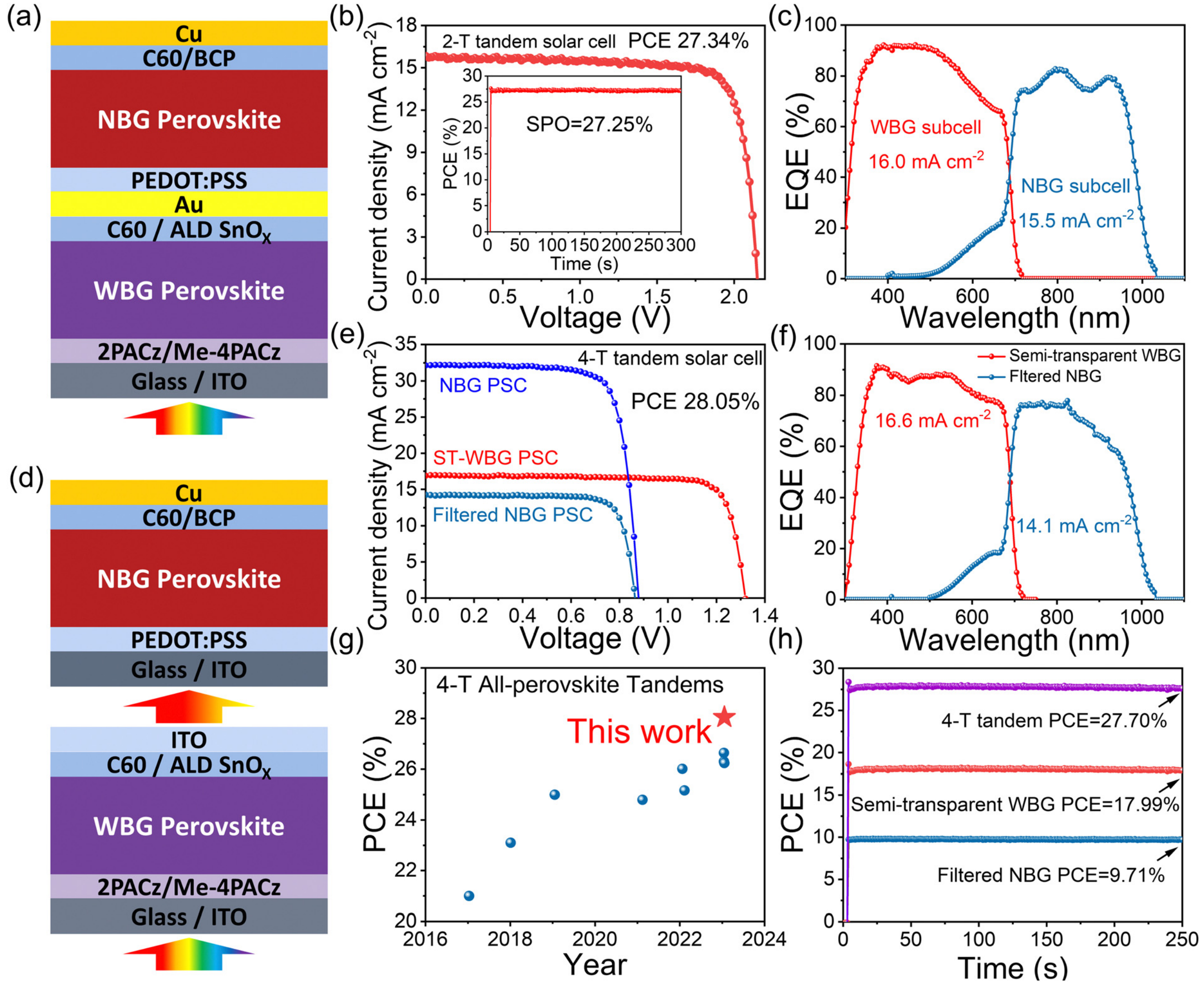

Fig. 5 (a) The device architecture of perovskite/perovskite 2-T tandem solar cells. (b) The *J–V* curves and steady-state power outputs measured at a bias voltage of 1.9 V of the champion 2-T tandem solar cell. (c) EQE curves of the best-performing 2-T tandem cell. (d) The device architecture of perovskite/perovskite 4-T tandem solar cells. (e) The *J–V* curves of three different devices: a single-junction NBG PSC, a single-junction semi-transparent WBG (ST-WGB) PSC, and a filtered NBG PSC. (f) EQE curves of the single-junction semi-transparent WBG PSC and the filtered NBG PSC. (g) The power conversion efficiency (PCE) evolution of the 4-T all-perovskite tandem solar cells. (h) The steady-state power outputs measured at bias voltages of 1.14 V and 0.74 V for the semi-transparent WBG PSC and the filtered NBG PSC, respectively.

Table 1 Summary of photovoltaic parameters of our champion devices of various types

Type	V_{OC} (V)	J_{SC} (mA cm^{-2})	FF (%)	PCE (%)
ST-WBG PSC	1.32	16.90	81.62	18.25
NBG PSC	0.88	32.14	77.41	21.83
Filtered NBG PSC	0.86	14.25	79.75	9.80
2-T tandem	2.15	15.81	80.49	**27.34**
4-T tandem				**28.05**

cells, as indicated in Fig. 5g and Table S4 (ESI†). To achieve more reliable device performance, the steady-state power output of the device was measured at the maximum power point (1.14 V for the semi-transparent WBG device and 0.74 V for the filtered NBG device) over 250 s (Fig. 5h). The stable output of the semi-transparent WBG device and the filtered NBG bottom cell reached stabilized PCE values of 17.99% and 9.71%, respectively, closely matching the values obtained from the *J–V* curve. These superior results indicate that the strategy of the Me-4PACz interface engineering improves both the stability and efficiency of WBG PSCs, further advancing the realization of efficient and stable all-perovskite tandem solar cells.

3. Conclusion

In conclusion, we have demonstrated the significant performance enhancements of WBG perovskite single-junction and tandem solar cells through the utilization of Me-4PACz interface engineering. By focusing on the interface of the hole selection layer and perovskite layer crystallization which is difficult to control and by introducing this novel approach, we have achieved remarkable improvements in various key parameters, including V_{OC}, FF, and PCE. The opaque and semi-transparent 1.78 eV-bandgap perovskite solar cells have exhibited impressive champion power conversion efficiencies of 19.83% and 18.25%, respectively. These results highlight the effectiveness of Me-4PACz interface engineering in enhancing the performance of WBG perovskite single-junction devices. Furthermore, we have also achieved similar improvements in 1.66 eV perovskite cells, making them suitable for perovskite/silicon tandem solar cell applications. Taking the advantage of this strategy, we were able to make efficient all-perovskite tandem solar cells. 2-T and 4-T tandem cells have achieved exceptional efficiencies of 27.34% and 28.05%, respectively, showcasing their potential for efficient multi-junction tandem solar cell applications. Overall, our investigation and regulation at the bottom interface of the perovskite layer provides valuable insights for further improvement and optimization in the field of perovskite photovoltaics.

4. Experimental section

Experimental details are provided in the ESI.†

Author contributions

Conceptualization, H. C.; methodology, H. C.; investigation, H. C.; formal analysis, H. C., L. H., S. Z., C. W., X. H., H. G., S. W., W. S., D. P., K. D., J. Z., and P. J.; funding acquisition, W. K. and G. F.; project administration, G. F.; resources, H. C.; validation, H. C.; visualization, H. C.; writing – original draft, H. C.; writing – review and editing, H. C., C. T., W. K., and G. F.; supervision, W. W., C. T., W. K., and G. F.

Data availability

The data that support the findings of this study are available in the ESI† of this article.

Conflicts of interest

The authors declare no conflict of interest.

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References

1 A. Kojima, K. Teshima, Y. Shirai and T. Miyasaka, *J. Am. Chem. Soc.*, 2009, **131**, 6050–6051.
2 S. D. Stranks, G. E. Eperon, G. Grancini, C. Menelaou, M. J. P. Alcocer, T. Leijtens, L. M. Herz, A. Petrozza and H. J. Snaith, *Science*, 2013, **342**, 341–344.
3 S. De Wolf, J. Holovsky, S.-J. Moon, P. Löper, B. Niesen, M. Ledinsky, F.-J. Haug, J.-H. Yum and C. Ballif, *J. Phys. Chem. Lett.*, 2014, **5**, 1035–1039.
4 X. Y. Chin, D. Cortecchia, J. Yin, A. Bruno and C. Soci, *Nat. Commun.*, 2015, **6**, 7383.
5 T. Jesper Jacobsson, J.-P. Correa-Baena, M. Pazoki, M. Saliba, K. Schenk, M. Grätzel and A. Hagfeldt, *Energy Environ. Sci.*, 2016, **9**, 1706–1724.
6 Y. Zhao, F. Ma, Z. Qu, S. Yu, T. Shen, H.-X. Deng, X. Chu, X. Peng, Y. Yuan, X. Zhang and J. You, *Science*, 2022, **377**, 531–534.
7 Q. Jiang, Y. Zhao, X. Zhang, X. Yang, Y. Chen, Z. Chu, Q. Ye, X. Li, Z. Yin and J. You, *Nat. Photonics*, 2019, **13**, 460–466.
8 J. J. Yoo, G. Seo, M. R. Chua, T. G. Park, Y. Lu, F. Rotermund, Y.-K. Kim, C. S. Moon, N. J. Jeon, J.-P. Correa-Baena, V. Bulović, S. S. Shin, M. G. Bawendi and J. Seo, *Nature*, 2021, **590**, 587–593.
9 Q. Tan, Z. Li, G. Luo, X. Zhang, B. Che, G. Chen, H. Gao, D. He, G. Ma, J. Wang, J. Xiu, H. Yi, T. Chen and Z. He, *Nature*, 2023, **613**, 545–551.
10 H. Chen, A. Maxwell, C. Li, S. Teale, B. Chen, T. Zhu, E. Ugur, G. Harrison, L. Grater, J. Wang, Z. Wang, L. Zeng, S. M. Park, L. Chen, P. Serles, R. A. Awni, B. Subedi, X. Zheng, C. Xiao, N. J. Podraza, T. Filleter, C. Liu, Y. Yang, J. M. Luther, S. De Wolf, M. G. Kanatzidis, Y. Yan and E. H. Sargent, *Nature*, 2023, **613**, 676–681.
11 R. Lin, J. Xu, M. Wei, Y. Wang, Z. Qin, Z. Liu, J. Wu, K. Xiao, B. Chen, S. M. Park, G. Chen, H. R. Atapattu, K. R. Graham, J. Xu, J. Zhu, L. Li, C. Zhang, E. H. Sargent and H. Tan, *Nature*, 2022, **603**, 73–78.
12 A. Al-Ashouri, E. Köhnen, B. Li, A. Magomedov, H. Hempel, P. Caprioglio, J. A. Márquez, A. B. Morales Vilches, E. Kasparavicius, J. A. Smith, N. Phung, D. Menzel, M. Grischek, L. Kegelmann, D. Skroblin, C. Gollwitzer, T. Malinauskas, M. Jošt, G. Matič, B. Rech, R. Schlatmann, M. Topič, L. Korte, A. Abate, B. Stannowski, D. Neher, M. Stolterfoht, T. Unold, V. Getautis and S. Albrecht, *Science*, 2020, **370**, 1300–1309.
13 P. Tockhorn, J. Sutter, A. Cruz, P. Wagner, K. Jäger, D. Yoo, F. Lang, M. Grischek, B. Li, J. Li, O. Shargaieva, E. Unger, A. Al-Ashouri, E. Köhnen, M. Stolterfoht, D. Neher, R. Schlatmann, B. Rech, B. Stannowski, S. Albrecht and C. Becker, *Nat. Nanotechnol.*, 2022, **17**, 1214–1221.
14 Q. Jiang, J. Tong, R. A. Scheidt, X. Wang, A. E. Louks, Y. Xian, R. Tirawat, A. F. Palmstrom, M. P. Hautzinger, S. P. Harvey, S. Johnston, L. T. Schelhas, B. W. Larson, E. L. Warren, M. C. Beard, J. J. Berry, Y. Yan and K. Zhu, *Science*, 2022, **378**, 1295–1300.
15 J. Liu, M. De Bastiani, E. Aydin, G. T. Harrison, Y. Gao, R. R. Pradhan, M. K. Eswaran, M. Mandal, W. Yan, A. Seitkhan, M. Babics, A. S. Subbiah, E. Ugur, F. Xu, L. Xu, M. Wang, A. U. Rehman, A. Razzaq, J. Kang, R. Azmi, A. A. Said, F. H. Isikgor, T. G. Allen, D. Andrienko, U. Schwingenschlögl, F. Laquai and S. D. Wolf, *Science*, 2022, **377**, 302–306.
16 G. Yang, Z. Ni, Z. J. Yu, B. W. Larson, Z. Yu, B. Chen, A. Alasfour, X. Xiao, J. M. Luther, Z. C. Holman and J. Huang, *Nat. Photonics*, 2022, **16**, 588–594.

17 Z. Li, X. Li, X. Chen, X. Cui, C. Guo, X. Feng, D. Ren, Y. Mo, M. Yang, H. Huang, R. Jia, X. Liu, L. Han, S. Dai and M. Cai, *Joule*, 2023, S254243512300168X.
18 T. Li, J. Xu, R. Lin, S. Teale, H. Li, Z. Liu, C. Duan, Q. Zhao, K. Xiao, P. Wu, B. Chen, S. Jiang, S. Xiong, H. Luo, S. Wan, L. Li, Q. Bao, Y. Tian, X. Gao, J. Xie, E. H. Sargent and H. Tan, *Nat. Energy*, 2023, **8**, 610–620.
19 R. He, W. Wang, Z. Yi, F. Lang, C. Chen, J. Luo, J. Zhu, J. Thiesbrummel, S. Shah, K. Wei, Y. Luo, C. Wang, H. Lai, H. Huang, J. Zhou, B. Zou, X. Yin, S. Ren, X. Hao, L. Wu, J. Zhang, J. Zhang, M. Stolterfoht, F. Fu, W. Tang and D. Zhao, *Nature*, 2023, **618**, 80–86.
20 J. Zhu, Y. Luo, R. He, C. Chen, Y. Wang, J. Luo, Z. Yi, J. Thiesbrummel, C. Wang, F. Lang, H. Lai, Y. Xu, J. Wang, Z. Zhang, W. Liang, G. Cui, S. Ren, X. Hao, H. Huang, Y. Wang, F. Yao, Q. Lin, L. Wu, J. Zhang, M. Stolterfoht, F. Fu and D. Zhao, *Nat. Energy*, 2023, **8**, 714–724.
21 Z. Li, X. Li, X. Chen, X. Cui, C. Guo, X. Feng, D. Ren, Y. Mo, M. Yang, H. Huang, R. Jia, X. Liu, L. Han, S. Dai and M. Cai, *Joule*, 2023, **7**, 1363–1381.
22 R. Lin, Y. Wang, Q. Lu, B. Tang, J. Li, H. Gao, Y. Gao, H. Li, C. Ding, J. Wen, P. Wu, C. Liu, S. Zhao, K. Xiao, Z. Liu, C. Ma, Y. Deng, L. Li, F. Fan and H. Tan, *Nature*, 2023, **620**, 994–1000.
23 Z. Liu, J. Siekmann, B. Klingebiel, U. Rau and T. Kirchartz, *Adv. Energy Mater.*, 2021, **11**, 2003386.
24 R. D. J. Oliver, P. Caprioglio, F. Peña-Camargo, L. R. V. Buizza, F. Zu, A. J. Ramadan, S. G. Motti, S. Mahesh, M. M. McCarthy, J. H. Warby, Y.-H. Lin, N. Koch, S. Albrecht, L. M. Herz, M. B. Johnston, D. Neher, M. Stolterfoht and H. J. Snaith, *Energy Environ. Sci.*, 2022, **15**, 714–726.
25 P. Caprioglio, J. A. Smith, R. D. J. Oliver, A. Dasgupta, S. Choudhary, M. D. Farrar, A. J. Ramadan, Y.-H. Lin, M. G. Christoforo, J. M. Ball, J. Diekmann, J. Thiesbrummel, K.-A. Zaininger, X. Shen, M. B. Johnston, D. Neher, M. Stolterfoht and H. J. Snaith, *Nat. Commun.*, 2023, **14**, 932.
26 J. Thiesbrummel, F. Peña-Camargo, K. O. Brinkmann, E. Gutierrez-Partida, F. Yang, J. Warby, S. Albrecht, D. Neher, T. Riedl, H. J. Snaith, M. Stolterfoht and F. Lang, *Adv. Energy Mater.*, 2023, **13**, 2202674.
27 K. Datta, J. Wang, D. Zhang, V. Zardetto, W. H. M. Remmerswaal, C. H. L. Weijtens, M. M. Wienk and R. A. J. Janssen, *Adv. Mater.*, 2022, **34**, 2110053.
28 M. Stolterfoht, C. M. Wolff, J. A. Márquez, S. Zhang, C. J. Hages, D. Rothhardt, S. Albrecht, P. L. Burn, P. Meredith, T. Unold and D. Neher, *Nat. Energy*, 2018, **3**, 847–854.
29 M. Grischek, P. Caprioglio, J. Zhang, F. Peña-Camargo, K. Sveinbjörnsson, F. Zu, D. Menzel, J. H. Warby, J. Li, N. Koch, E. Unger, L. Korte, D. Neher, M. Stolterfoht and S. Albrecht, *Sol. RRL*, 2022, **6**, 2200690.
30 P. Hang, C. Kan, B. Li, Y. Yao, Z. Hu, Y. Zhang, J. Xie, Y. Wang, D. Yang and X. Yu, *Adv. Funct. Mater.*, 2023, **33**, 2214381.
31 Q. Xu, B. Shi, Y. Li, L. Yan, W. Duan, Y. Li, R. Li, N. Ren, W. Han, J. Liu, Q. Huang, D. Zhang, H. Ren, S. Xu, C. Zhang, H. Zhuang, A. Lambertz, K. Ding, Y. Zhao and X. Zhang, *Adv. Energy Mater.*, 2022, **12**, 2202404.
32 C. Wang, W. Shao, J. Liang, C. Chen, X. Hu, H. Cui, C. Liu, G. Fang and C. Tao, *Small*, 2022, **18**, 2204081.
33 R. He, Z. Yi, Y. Luo, J. Luo, Q. Wei, H. Lai, H. Huang, B. Zou, G. Cui, W. Wang, C. Xiao, S. Ren, C. Chen, C. Wang, G. Xing, F. Fu and D. Zhao, *Adv. Sci.*, 2022, **9**, 2203210.
34 S. G. Ji, I. J. Park, H. Chang, J. H. Park, G. P. Hong, B. K. Choi, J. H. Jang, Y. J. Choi, H. W. Lim, Y. J. Ahn, S. J. Park, K. T. Nam, T. Hyeon, J. Park, D. H. Kim and J. Y. Kim, *Joule*, 2022, **6**, 2390–2405.
35 J. Liang, C. Chen, X. Hu, Z. Chen, X. Zheng, J. Li, H. Wang, F. Ye, M. Xiao, Z. Lu, Y. Xu, S. Zhang, R. Yu, C. Tao and G. Fang, *ACS Appl. Mater. Interfaces*, 2020, **12**, 48458–48466.
36 H. Lai, J. Luo, Y. Zwirner, S. Olthof, A. Wieczorek, F. Ye, Q. Jeangros, X. Yin, F. Akhundova, T. Ma, R. He, R. K. Kothandaraman, X. Chin, E. Gilshtein, A. Müller, C. Wang, J. Thiesbrummel, S. Siol, J. M. Prieto, T. Unold, M. Stolterfoht, C. Chen, A. N. Tiwari, D. Zhao and F. Fu, *Adv. Energy Mater.*, 2022, **12**, 2202438.
37 N. Yan, Y. Gao, J. Yang, Z. Fang, J. Feng, X. Wu, T. Chen and S. (Frank) Liu, *Angew. Chem., Int. Ed.*, 2023, **62**, e202216668.
38 L. Tian, R. Tyburski, C. Wen, R. Sun, M. Abdellah, J. Huang, L. D'Amario, G. Boschloo, L. Hammarström and H. Tian, *J. Am. Chem. Soc.*, 2020, **142**, 18668–18678.
39 C. C. Boyd, R. C. Shallcross, T. Moot, R. Kerner, L. Bertoluzzi, A. Onno, S. Kavadiya, C. Chosy, E. J. Wolf, J. Werner, J. A. Raiford, C. de Paula, A. F. Palmstrom, Z. J. Yu, J. J. Berry, S. F. Bent, Z. C. Holman, J. M. Luther, E. L. Ratcliff, N. R. Armstrong and M. D. McGehee, *Joule*, 2020, **4**, 1759–1775.
40 P. Ru, E. Bi, Y. Zhang, Y. Wang, W. Kong, Y. Sha, W. Tang, P. Zhang, Y. Wu, W. Chen, X. Yang, H. Chen and L. Han, *Adv. Energy Mater.*, 2020, **10**, 1903487.
41 F. H. Isikgor, S. Zhumagali, L. V. T. Merino, M. De Bastiani, I. McCulloch and S. D. Wolf, *Nat. Rev. Mater.*, 2022, **8**, 89–108.
42 L. Mao, T. Yang, H. Zhang, J. Shi, Y. Hu, P. Zeng, F. Li, J. Gong, X. Fang, Y. Sun, X. Liu, J. Du, A. Han, L. Zhang, W. Liu, F. Meng, X. Cui, Z. Liu and M. Liu, *Adv. Mater.*, 2022, **34**, 2206193.
43 A. Farag, T. Feeney, I. M. Hossain, F. Schackmar, P. Fassl, K. Küster, R. Bäuerle, M. A. Ruiz-Preciado, M. Hentschel, D. B. Ritzer, A. Diercks, Y. Li, B. A. Nejand, F. Laufer, R. Singh, U. Starke and U. W. Paetzold, *Adv. Energy Mater.*, 2023, **13**, 2203982.
44 W. Peng, K. Mao, F. Cai, H. Meng, Z. Zhu, T. Li, S. Yuan, Z. Xu, X. Feng, J. Xu, M. D. McGehee and J. Xu, *Science*, 2023, **379**, 683–690.
45 B. Niu, H. Liu, Y. Huang, E. Gu, M. Yan, Z. Shen, K. Yan, B. Yan, J. Yao, Y. Fang, H. Chen and C. Li, *Adv. Mater.*, 2023, 2212258.
46 L. Li, Y. Wang, X. Wang, R. Lin, X. Luo, Z. Liu, K. Zhou, S. Xiong, Q. Bao, G. Chen, Y. Tian, Y. Deng, K. Xiao, J. Wu,

M. I. Saidaminov, H. Lin, C.-Q. Ma, Z. Zhao, Y. Wu, L. Zhang and H. Tan, *Nat. Energy*, 2022, **7**, 708–717.
47 W. Wang, X. Liu, J. Wang, C. Chen, J. Yu, D. Zhao and W. Tang, *Adv. Energy Mater.*, 2023, 2300694.
48 L. Li, Y. Wang, X. Wang, R. Lin, X. Luo, Z. Liu, K. Zhou, S. Xiong, Q. Bao, G. Chen, Y. Tian, Y. Deng, K. Xiao, J. Wu, M. I. Saidaminov, H. Lin, C.-Q. Ma, Z. Zhao, Y. Wu, L. Zhang and H. Tan, *Nat. Energy*, 2022, **7**, 708–717.
49 Y. Deng, X. Zheng, Y. Bai, Q. Wang, J. Zhao and J. Huang, *Nat. Energy*, 2018, **3**, 560–566.
50 P. Caprioglio, M. Stolterfoht, C. M. Wolff, T. Unold, B. Rech, S. Albrecht and D. Neher, *Adv. Energy Mater.*, 2019, **9**, 1901631.
51 S. Chen, X. Dai, S. Xu, H. Jiao, L. Zhao and J. Huang, *Science*, 2021, **373**, 902–907.
52 C. Bi, Q. Wang, Y. Shao, Y. Yuan, Z. Xiao and J. Huang, *Nat. Commun.*, 2015, **6**, 7747.
53 Z. Li, B. Li, X. Wu, S. A. Sheppard, S. Zhang, D. Gao, N. J. Long and Z. Zhu, *Science*, 2022, **376**, 416–420.
54 C. Ma, M.-C. Kang, S.-H. Lee, S. J. Kwon, H.-W. Cha, C.-W. Yang and N.-G. Park, *Joule*, 2022, **6**, 2626–2643.
55 C. Zhu, X. Niu, Y. Fu, N. Li, C. Hu, Y. Chen, X. He, G. Na, P. Liu, H. Zai, Y. Ge, Y. Lu, X. Ke, Y. Bai, S. Yang, P. Chen, Y. Li, M. Sui, L. Zhang, H. Zhou and Q. Chen, *Nat. Commun.*, 2019, **10**, 815.
56 J. S. Kim, J.-M. Heo, G.-S. Park, S.-J. Woo, C. Cho, H. J. Yun, D.-H. Kim, J. Park, S.-C. Lee, S.-H. Park, E. Yoon, N. C. Greenham and T.-W. Lee, *Nature*, 2022, **611**, 688–694.
57 X. Hu, J. Li, C. Wang, H. Cui, Y. Liu, S. Zhou, H. Guan, W. Ke, C. Tao and G. Fang, *Nano-Micro Lett.*, 2023, **15**, 103.